Ashutosh Thapa

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Education

University of Louisiana at Monroe — Monroe, LA
Bachelor of Science in Computer Science — *Expected May 2027*

- **Coursework**: Data Structures, Operating Systems, Computer Architecture, Database Management (MySQL on AWS EC2), Discrete Structures, Mathematical Statistics

Projects

Tripwire | *Python, AWS Lambda, EventBridge, DynamoDB, Bedrock, Terraform* — In Progress

- Detected 9 of 9 events in a simulated stolen-key attack at under 0.3% false positives across 5,000 normal CloudTrail events, by scoring each API call with a per-identity Isolation Forest on 7 rarity features
- Reduced each incident to one Slack alert by grouping events by identity and source IP, mapping them to MITRE ATT&CK techniques, and having Amazon Bedrock write the summary and the key to rotate
- Made the full stack reproducible with one `terraform apply` by defining every resource as code and replaying Stratus Red Team attacks in GitHub Actions

HawkPlan | *Next.js, TypeScript, PostgreSQL, pgvector, Python, RAG* — In Progress

- Turning ULM's catalog into a degree planner that outputs a valid semester-by-semester schedule for every CS track, by parsing courses into a prerequisite graph and ordering it with topological sort under credit-hour limits
- Answering advising questions with cited catalog sections, measured on a 50-question test set from real advising FAQs, by building a RAG pipeline over the catalog with pgvector

Satya | *PyTorch, Hugging Face, ONNX, FastAPI, Docker, AWS ECS* — In Progress

- Selecting a fake news classifier by per-class precision, recall, and macro F1 on LIAR and FakeNewsNet, by fine-tuning and comparing DistilBERT and RoBERTa on the same splits
- Shrinking model weights about 4x for CPU-only serving by exporting to ONNX with INT8 quantization behind a FastAPI container on AWS ECS
- Labeling posts in the feed with one API request per scroll instead of one per post, by batching visible posts in a Chrome extension

Relay | *TypeScript, Node.js, Redis, PostgreSQL, Docker, k6* — In Progress

- Cutting repeat LLM spend, tracked as cache hit rate and dollars saved per day, by returning stored answers for prompts above 0.95 cosine similarity from a Redis semantic cache
- Keeping client apps up during provider outages, verified by k6 load tests with injected 503 errors, by failing over between providers with per-key token bucket rate limits

Experience

Volunteer Web and IT Assistant — Jan 2020 – Dec 2021
Jamnapur Helped Society — *Chitwan, Nepal*

- Raised visibility for earthquake relief fundraising by building and maintaining the organization's website and producing animated appeal videos
- Cut printing and admin costs by moving paper and spreadsheet donor records into organized cloud storage the whole team could search
- Kept relief drives running by setting up and troubleshooting staff computers and coordinating donation logistics

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Web and Backend: React, Next.js, Node.js, FastAPI, REST APIs, PostgreSQL, MySQL, Redis
AI/ML: PyTorch, Hugging Face Transformers, scikit-learn, pandas, Amazon Bedrock, RAG, pgvector, ONNX
Cloud and Tools: AWS (EC2, S3, Lambda, IAM, CloudTrail, ECS), Terraform, Docker, Git, GitHub Actions, Linux
Certifications: AWS Certified AI Practitioner, AWS Certified Cloud Practitioner, EC-Council Ethical Hacking Essentials (EHE)